

Batch: A3 Roll No.: 16010324044
Experiment / assignment / tutorial No. 6
Grade: AA / AB / BB / BC / CC / CD / DD
Signature of the Staff In-charge with date

TITLE: Static Routing protocol using Cisco Hardware.

AIM: To configure Static Routing protocol for given network topology using Cisco Hardware.

OUTCOME: Student will be able to analyze various routing techniques and design complete network as per the given requirements

Theory:

Routers forward packets using either route information from route table entries that you manually configure or the route information that is calculated using dynamic routing algorithms.

Static routes, which define explicit paths between two routers, cannot be automatically updated; the administrator must manually reconfigure static routes when network changes occur. Static routes use less bandwidth than dynamic routes. No CPU cycles are used to calculate and analyze routing updates. The network administrator can supplement dynamic routes with static routes where appropriate. We can redistribute static routes into dynamic routing algorithms, but we cannot redistribute routing information calculated by dynamic routing algorithms into the static routing table.

Static routes are preferred in environments where network traffic is predictable and where the network design is simple. Static routes are not used in large, constantly changing networks because static routes cannot react to network changes. Most networks use dynamic routes to communicate between routers but might have one or two static routes configured for special cases. Static routes are also useful for specifying a gateway of last resort (a default router to which all unroutable packets are sent).

Static routes have a default administrative distance of 1. A router prefers a static route to a dynamic route because the router considers a route with a low number to be the shortest.

Directly Connected Static Routes

Here we specify only the output interface (the interface on which all packets are sent to the destination network) in a directly connected static route. The router assumes the destination is directly attached to the output interface and the packet destination is used as the next-hop address. The next hop can be an interface, only for point-to-point interfaces. For broadcast interfaces, the next hop must be an IPv4/IPv6 address.

Fully Specified Static Routes

Here we must specify either the output interface (the interface on which all packets are sent to the destination network) or the next-hop address in a fully specified static route. A fully

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specified static route can be used when the output interface is a multi-access interface and you need to identify the next-hop address. The next-hop address must be directly attached to the specified output interface.

Floating Static Routes

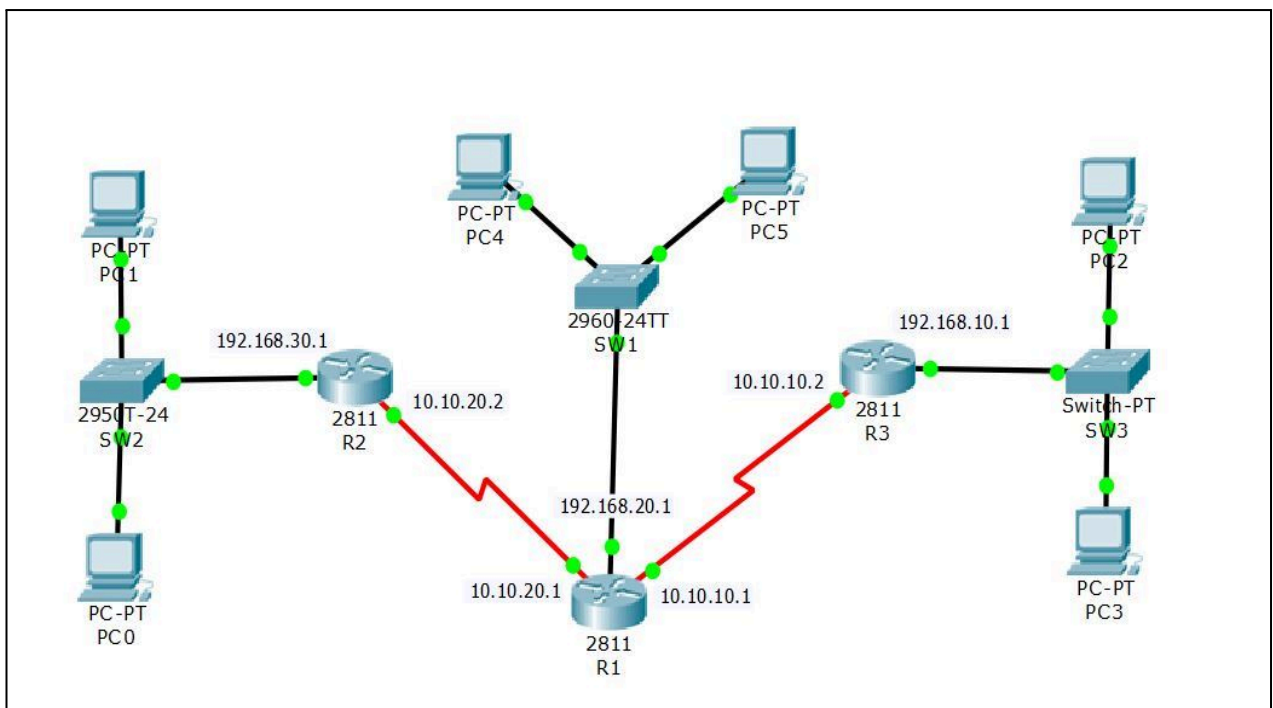
A floating static route is a static route that the router uses to back up a dynamic route. You must configure a floating static route with a higher administrative distance than the dynamic route that it backs up. In this instance, the router prefers a dynamic route to a floating static route. Floating static route can be used as a replacement if the dynamic route is lost.

Procedure:

1. To begin with the experiment, open Cisco Packet Tracer
2. Create the lab and configure the basic IP addressing for all PCs and Routers
3. Using the command line interface of the router or switch enter the global configuration mode of router using command **configure terminal**
4. Static Routing command syntax: **ip route [destination network address] [mask] [next hop address or interface name]**
5. Save the configuration using the command **copy running-config startup-config**
6. Observe the routes generated in the routing table of the router for the given network topology using command **show ip route**
7. Using **ping** command test the ability of the source computer to reach the destination computer

Example:

Create a network topology using cisco packet tracer and configure static routes on the router using command line interface.



```
COM3 - PuTTY

User Access Verification
Username: admin
Password:
Rack2_R1>
*Sep 30 10:32:53.289: %SEC_LOGIN-5-LOGIN_SUCCESS: Login Success [user: admin] [Source: UNKNOWN] [localport: 0] at 10:32:53 UTC Tue Sep 30 2025enable
Password:
Rack2_R1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Rack2_R1(config)#do show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0     unassigned      YES NVRAM   administratively down down
GigabitEthernet0/0/1     unassigned      YES NVRAM   administratively down down
GigabitEthernet0/1/0     unassigned      YES NVRAM   administratively down down
Serial0/2/0              unassigned      YES NVRAM   down         down
Serial0/2/1              unassigned      YES NVRAM   down         down
Rack2_R1(config)#interface gig 0/0/0
Rack2_R1(config-if)#ip address 20.0.0.1
% Incomplete command.

Rack2_R1(config-if)#ip address 20.0.0.1 255.255.255.0
Rack2_R1(config-if)#no shutdown
Rack2_R1(config-if)#
*Sep 30 10:39:11.093: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to down
Rack2_R1(config-if)#exit
```

```
COM3 - PuTTY

Rack2_R1(config-if)#exit
Rack2_R1(config)#interface 0/0/1
^
% Invalid input detected at '^' marker.

Rack2_R1(config)#interface gig 0/0/1
Rack2_R1(config-if)#ip address 10.0.0.1
% Incomplete command.

Rack2_R1(config-if)#ip address 10.0.0.1 255.255.255.0
Rack2_R1(config-if)#no shutdown
Rack2_R1(config-if)#ex
*Sep 30 10:40:17.468: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to down
Rack2_R1(config)#exit
Rack2_R1#
*Sep 30 10:40:23.523: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to up
*Sep 30 10:40:23.613: %SYS-5-CONFIG I: Configured from console by admin on console
*Sep 30 10:40:24.523: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
*Sep 30 10:42:47.531: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/0, changed state to up
*Sep 30 10:42:48.531: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
Rack2_R1#do show ip interface brief
^
% Invalid input detected at '^' marker.

Rack2_R1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Rack2_R1(config)#do show config terminal brief
^
% Invalid input detected at '^' marker.

Rack2_R1(config)#do show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0     20.0.0.1        YES manual up          up
GigabitEthernet0/0/1     10.0.0.1        YES manual up          up
GigabitEthernet0/1/0     unassigned      YES NVRAM   administratively down down
Serial0/2/0              unassigned      YES NVRAM   down         down
Serial0/2/1              unassigned      YES NVRAM   down         down
Rack2_R1(config)#ip route 40.0.0.0 255.255.255.0 20.0.0.2
Rack2_R1(config)#ip route 30.0.0.0 255.255.255.0 20.0.0.2
Rack2_R1(config)#ip route 50.0.0.0 255.255.255.0 20.0.0.2
Rack2_R1(config)#
*Sep 30 11:02:45.601: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to down
*Sep 30 11:02:46.601: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to down
*Sep 30 11:02:52.601: %LINK-3-UPDOWN: Interface GigabitEthernet0/0/1, changed state to up
*Sep 30 11:02:53.601: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
```

CONCLUSION

In this experiment, I successfully configured static routing between two routers using Cisco Packet Tracer. The router interfaces were assigned appropriate IP addresses and static routes were configured manually using the ip route command. The routing tables were verified using the show ip route command, and successful communication between PCs on different networks was confirmed using the ping command. Thus, the working of static routing and manual route configuration was successfully demonstrated.

Signature of faculty in-charge

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